

OCCULT TOXIC STATES OF APPARENTLY NON-TOXIC PATIENTS, MAINTAINED ON PROLONGED LOW DOSE METHOTREXATE WITHOUT RESCUE

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ABSTRACT

Systemic chemotherapy is now, an important part of cancer treatment. Methotrexate, a folic acid analogue, is the most commonly used antimetabolite, for this purpose. As most of clinical protocols based on this drug, are studied, and are produced in the West, so they cannot be applied efficiently and produces more toxicities in the patients in developing countries like Pakistan. The toxic states of the patients, who were maintained on Methotrexate since long, without rescue, in a particular dosage regimens, were evaluated according to WHO criteria. All of them had no apparent toxicities. Fifty percent of the patients had hemoglobin of grade I, II and III severity. Eight percent of patients had grade I leukopenia. Twenty percent of the patients had high urea levels of grade I and forty percent of the patients had high urea levels of grade I and forty percent of the them had high creatinine levels of grade I severity. All patients had severe hepatotoxicity, more than half had SGPT levels of grade II and alkaline phosphatase of grade III severity. The different parameters were also considered according to sex, dosage schedules, duration of therapy and total dose. Less haematological and renal but more hepatic toxicities, were present in female. Low dose was found to be more toxic than high dose. Longer the duration lowers the haematological but highers the renal and hepatic toxicities.

INTRODUCTION

System chemotherapy has become an important part of cancer therapy. The use of drugs either alone or as adjuvant to surgery or irradiation has led to significant advances in the cancer therapy. Antimetabolites have shown a definite role in this regard (Faber S. *et al.*, 1947; Segger (Faber S. *et al.*, 1947; Segger D.R. *et al.*, 1947; Smith J.M. *et al.*, 1948; Borsa & Whitamore 1969; Bertino J.R., 1971, 1975).

Methotrexate is a folic acid analogue and is the most commonly used antimetabolite. It competes with folic acid for the enzyme dihydrofolate reductase Bertino J.R. *et al.*, 1964), blocks the formation of tetrahydrofolic acid compounds and thus inhibit purines and pyrimidines to stop DNA synthesis and cell

division (Chabner B.A. *et al.*, 1958 and Skipper H.T. & Schabel F.M. Jr. 1982).

Methotrexate has shown dramatic results, both in combination and as a single agent in cancer therapy. In the treatment of acute lymphoblastic leukemia of childhood, for both the systemic and central nervous components, methotrexate has an effective role (Henderson E.S., 1973; Nagamoto A. & Sullivan M.P., 1973; Djerassi I., *et al.*, 1968). It is commonly used drug in the treatment of lymphomas, osteogenic sarcoma Jaffe N. *et al.*, 1974), squamous cell carcinoma of head and neck (Levit M. *et al.*, 1974) and breast cancer Bonadonna G. *et al.*, 1976). It is considered to produce objective responses in some patients with lung cancer, epidermoid carcinoma of cervix and a few other solid tumors.

Table-1
Characteristic of patients

No. of Patients	-	26
Mean age	-	57.33 years
Sex	-	Male-15, Female-II
Site of lesion	-	Head and Neck
Karnophsky score	-	Between 6—90
Average duration of methotrexate Treatment	-	6.37 Months
Average total	-	412.50 mg
Minimum drug Free interval	-	15 days

Table 2
Distribution of toxicities in drug treated group According to WHO criteria

Toxicity	Profile	Grade 0	Grade I	Grade II	Grade III	Grade IV
	Hb.	45.8% 11*	41.66% 10	4.1% 01	8.33% 02	00% 00
Haemat-	RBC	100.0% 25	00% 00	00% 00	00% 00	00% 00
ological	TLC	92.0% 23	8.00% 02	00% 00	00% 00	00% 00
	Platelets	100.0% 25	00% 00	00% 00	00% 00	00% 00
Renal	Urea	80.0% 20	20.00% 00	00% 00	00% 00	00% 00
	Creatinine	60.0% 15	40.00% 10	00% 00	00% 00	00% 00
SGPT	Alkaline	00% 00	11.50% 03	53.84% 14	34.60% 09	00% 00
Hepatic	Phos- phatase	00% 00	00% 00	7.60% 02	57.60% 15	34.6% 09
	Bilirubin	100.0% 24	00% 00	00% 00	00% 00	00% 00

Table 3
Comparison of toxicities in male female patients

Toxicity	Profile	Male	Female
	RBC ($\times 10^6/\text{cm}^3$)	43.70% (07/16) (less)*	20.00% (02/10) (less)
	TLC ($\times 10^3/\text{cm}^3$)	43.70% (07/16) (less)	40.00% (04/10) (less)
	Platelets ($\times 10^3/\text{cm}^3$)	00.00% (16/16) -	00.00% (00/10)
Haematological	Hb. (gm%)	71.40% (10/14) (less)	80.00% (08/10) (less)
	MCV (fL)	71.40% (10/14) (high)	60.00% (06/10) (high)
	MCH (pg)	100.00% (15/15) (less)	100.00% (10/10) (less)
	MCHC (gm%)	100.00% (15/15) (less)	100.00% (10/10) (less)
Renal	Urea (mg%)	33.33% (05/15) (high)	20.00 (02/10) (high)
	Creatinine (mg%)	86.66% (13/15) (high)	50.00% (05/10) (high)
	SGPT (μL)	100.00% (15/15) (high)	100.00% (10/10) (high)
Hepatic	Alkaline Phosphatase (μL)	100.00% (15/15) (high)	100.00% (10/10) (high)
	Bilirubin (mg%)	00.00% (00/15) -	40.00% (04/10) (high)

*Percentage of patients showing toxicity (No. of patients showing toxicity/ total No. of patients) Mode.

Over the last twenty to thirty years, the treatment schedules based on Methotrexate, have become numerous, varied and sometimes more complicated. It is not possible to offer a simple, clear cut guideline that any clinician might follow and be certain of achieving a particular result in a malignancy. As the existing treatment

schedules are not applicable in the form, in which they are developed and do not produce the results which are expected, due to a number of factors, it, therefore becomes essential to work out a new plan. Similarly it is equally important to evaluate the toxicities of the commonly used protocols in our patients, who have a different body status¹⁶,

than, those western patients, on whom these protocols have been designed. In this study, we have evaluated the toxicities in our patients, who were maintained on prolong Methotrexate therapy, in a particular dosage regimen.

MATERIALS AND METHODS

Twenty six patients were randomly selected, who were on prolong Methotrexate therapy. All had head and neck malignancies. These patients had tumor or ulcerative lesions on cheek, tongue, gums, palate, tonsils, laryngeal structures or pharynx. The histopathology of lesions of twenty one

patients was squamous cell carcinoma, while of two patients was dysplasia. Remaining three patients had squamous cell carcinoma clinically. They had received radiations and were on Methotrexate therapy due to residue or recurrence.

The physical well being of these patients was judged on karnofsky scale. The apparent toxicities were noted according to Who criteria of drug toxicities (Miller A.B. *et al.*, 1981). The patients were free from toxicities like loss of hair, loss of consciousness, fever, oral ulceration, nausea and vomiting, allergy, cutaneous manifestations, haemorrhage, pain, neurotoxicity, haematuria, diarrhoea, consti-

Table 4
Effect of dosage schedules on toxicities

Toxicity	Profile	Total dose less than 75 mg/month	Total dose less than 75 mg/month
	RBC ($\times 10^6$)	27.27% (03/11) (less)	33.33% (04/12) (less)
	TLC ($\times 10^3$)	54.54% (06/11) (less)	41.66% (05/12) (less)
	Platelets ($\times 10^3$)	00.00% (00) -	00.00% (00/12)
Haematological	Hb. (g%)	100.00% (10/10) (less)	50.00% (06/12) (less)
	MCV (FL)	60.00% (06/10) (high)	50.00% (06/12) (less)
	MCH (pg)	100.00% (10/10) (less)	100.00% (12/12) (less)
	MCHC (g%)	100.00% (10/10) (less)	100.00% (12/12) (less)
Renal	Urea (mg%)	33.33% (04/12) (high)	23.07 (03/13) (high)
	Creatinine (mg%)	75.00% (09/12) (high)	84.61% (11/13) (high)
	SGPT (μ L)	100.00% (12/12) (high)	100.00% (12/12) (high)
Hepatic	Alkaline Phosphatase (μ L)	100.00% (12/12) (high)	100.00% (12/12) (high)
	Bilirubin (mg%)	00.00% (00/12) -	33.33% (04/12) (high)

*Percentage of patients showing toxicity (No. of patients showing toxicity/ Total No. of patients) Mode.

pation and frank infection. The important informations about their treatment were obtained from their hospital medical records.

About seven c.c. of blood was collected from each patient, from a prominent cubital vein, by disposable syringe. The collection

Bilirubin, SGPT and Alkaline phosphatase were assessed by specific Merck kits on autoanalyzer, within twenty four hours of collection.

Thirty controls were also randomly selected. The blood samples were collected

Table 5
Effect of total duration of therapy on toxicities

Toxicity	Profile	Less than 6 months therapy	More than 6 months therapy
	RBC ($\times 10^6$)	27.27% (03/11) (less)	33.33% (04/12) (less)
	TLC ($\times 10^3$)	54.54% (06/11) (less)	41.66% (05/12) (less)
	Platelets ($\times 10^3$)	00.00% (00) -	00.00%
Haematological	Hb. (g%)	100.00% (10/10) (less)	50.00% (06/12) (less)
	MCV (FL)	100.00% (10/10) (less)	50.00% (06/12) (less)
	MCH (pg)	100.00% (10/10) (less)	100.00% (12/12) (less)
	MCHC (g%)	100.00% (10/10) (less)	100.00% (12/12) (less)
Renal	Urea (mg%)	15.38% (02/13) (high)	41.66% (05/12) (high)
	Creatinine (mg%)	84.60% (11/13) (high)	66.66% (08/12) (high)
	SGPT (μ L)	100.00% (12/12) (high)	100.00% (10/10) (high)
Hepatic	Alkaline Phosphatase (μ L)	100.00% (12/12) (high)	100.00% (12/12) (high)
	Bilirubin (mg%)	25.00% (03/12) (high)	41.66% (05/12) (high)

*Percentage of patients showing toxicity (No. of patients showing toxicity/ Total No. of patients) Mode.

was about fourteen days after previous Methotrexate intravenous injection. Each sample was divided into two glass bottles, 2 c.c. in one having equal amount of heparin and 5 c.c. in other without any anticoagulant and were used to determine important haematological and other parameters respectively.

Renal parameters viz. Urea and creatinine and Hepatic parameters viz.

and treated in the same way, to determine the control values.

Student 't' test was applied for statistical analysis of the results (Bailey N.T.J., 1976 and Gupta S.C. & Kapoor V.K., 1985).

RESULTS

Table 1 shows general characteristics of the patients.

Table 2 presents distribution of toxicities in drug treated group, according to WHO grading. In case of haematological toxicities, about half the patients had hemoglobin of grade I, II and III severity and 8% of grade

males had more haematological and renal toxicities. However both the sexes had 100% high SGPT and alkaline phosphatase. No male but 40% female patients had high Bilirubin levels. Table 4 Shows effect of

Table 6
Effect of total dose on toxicities

Toxicity	Profile	Less than 400 mg	More than 400 mg
	RBC ($\times 10^6$)	33.33% (04/12) (less)*	38.46% (05/13) (less)
	TLC ($\times 10^3$)	41.66% (05/12) (less)	46.15% (06/13) less)
	Platelets ($\times 10^3$)	00.00% (00) -	00.00% (00/13 (less))
Haematological	Hb. (g%)	54.54% (06/11) (less)	91.66% (11/12) (less)
	MCV (FL)	54.54% (06/11) (high)	58.33% (07/12) (high)
	MCH (pg)	100.00% (11/11) (less)	100.00% (11/11) (less)
	MCHC (g%)	100.00% (11/11) (less)	100.00% (11/11) (less)
Renal	Urea (mg%)	30.76% (04/13) (high)	25.00% (03/12) (high)
	Creatinine (mg%)	69.23% (09/13) (high)	83.33% (10/12) (high)
	SGPT (μ L)	100.00% (12/12) (high)	100.00% (12/12) (high)
Hepatic	Alkaline Phosphatase (μ L)	100.00% (12/12) (high)	100.00% (12/12) (high)
	Bilirubin (mg%)	38.40% (05/13) (high)	23.07% (03/13) (high)

*Percentage of patients showing toxicity (No. of patients showing toxicity/ Total No. of patients) Mode.

I leukopenia. In case of renal toxicities, 20% patients had high urea of grade I and 40% had high creatinine of grade I toxic level. All patients had severe hepatotoxicity, more than half of the patients had SGPT levels of grade II and alkaline phosphatase of grade III severity.

Table 3 compares different toxicities in male and female patients and it is clear that

dosage schedules on toxicities. Less dose produced more haematological and renal toxicities, than high dose. Although hepatic toxicity was same in both the groups, but a high Bilirubin level was found in 33.33% patients receiving a total dose more than 75 mg/months.

Table 5 shows effect of total duration of therapy on toxicities. The patients who had

Methotrexate therapy less than 6 months presented more haematological but less renal.

Had hepatic toxicities which reverse in true for the patients who had Methotrexate therapy more than 6 months.

Table 6 compares the different toxicities in patients who had received total drug less or more than 400 mg upto their entry into this study. The patients who had received a total dose, more than 400 mg had more haematological but less renal and hepatic toxicities, while the patients who had a total dose less than 400 mg had less haematological but more renal and hepatic toxicities.

DISCUSSION

Experienced investigators have demonstrated that, schedules based upon Methotrexate, must be designed to suit specific type of cancer, regimens must be adjusted to individual patient response, the body characteristics, nutritional status, vital organ functions and so on. Therefore even, in the presence of a number of protocols, we can not offer, the same schedules, to our patients, which have been evaluated in western hospitals and we can not be assured of same benefits in our patients, which have been claimed by different oncology groups, in developed countries.

Livingstone and Carter have reviewed systemic Methotrexate therapy in head and neck cancers (Livingstone R.B. & Carter S.K., 1970). They showed that in 232 evaluable cases, tumor regression of more than 50% was achieved in 37.5% of cases overall, the range was 20% to 57%. Complete remissions were exceptions rather than the rule and benefits lasted between one to five months. IN almost all of these studies, the efficiency of Methotrexate was compromised to one degree or another by sometimes unexpected and apparently quite sudden severa toxicity, resulting directly from drug use. Now two facts stem out. First in respect

of the response of the disease, no reliable protocol is available, second in regard of the toxicities of the therapy a better dosage regime still requires exploration.

Recurrence in head and neck cancers is treated by a simple protocol in our centre. It consists of 50 mg Methotrexate I.V every two weeks up to nine doses. The drug is continued for further doses, if the disease is stationary or aggressive in absence of the drug. This simple protocol is according to our strategies. It suits to individual factors like body characteristics, nutritional status, vital organ functions of most of our patients. It is well applicable in our busy OPD setting, lack of facilities for inter-dosage parametry, lack of availability of drug and trained staff. It is also possible to continue the drug therapy according to this simple schedule even in the presence of poor compliance of patients, low economics, lack of interest of their families in the treatment ad their co-operation to follow the treatment.

The patients, free from gross toxicities were randomly selected to evaluate haematological, hepatic and renal functions. The toxicities were noted after 14 days of last dose administration, so these can be minimized by boy's improving mechanism, as much as possible and these toxicities were supposed to be permanent. They had hypochromic macrocytic anemia, of less severity, which may be due to poor nutritions of the patients instead of direct effect of the drug on myloid tissue, as expected in head and neck malignancies. Their renal toxicities were also not considerable. However they had hepatic toxicities of alarming limits. This requires a more detail hepatic functions evaluation.

The effects of different factors on toxicities is also important. It may help to understand nature of toxic reactions and to find out some detoxifying agent. It is evident that.

- (i) there is less haematological and renal but more hepatic toxicities in female

while a reverse pattern is seen in male. This probably a result of different hormonal status.

- (ii) A low dose is more toxic than a high dose. A fact which has already been proved by other investigators (Goldie J.H., *et al.*, 1972).
- (iii) Haematological toxicities decrease (development of tolerance) while, renal and hepatic toxicities increase with the passage of time. Again, a fact which has been already noted by a number of investigators (Dahl M.G.C. *et al.*, 1972).

Total amount of drug is also important as less amount produces more haematological but less renal and hepatic toxicities while high amount induces less haematological but more renal and hepatic toxicities.

This study is a preliminary survey of multisystems to find out the relative involvement of different organs and tissues of the body in Methotrexate induced toxicities. However further evaluation of other parameters of a system to understand the nature and locus of involvement, is highly recommended.

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